

Unit - IV Types of Learning

4.1 Types of Learning

Supervised Learning

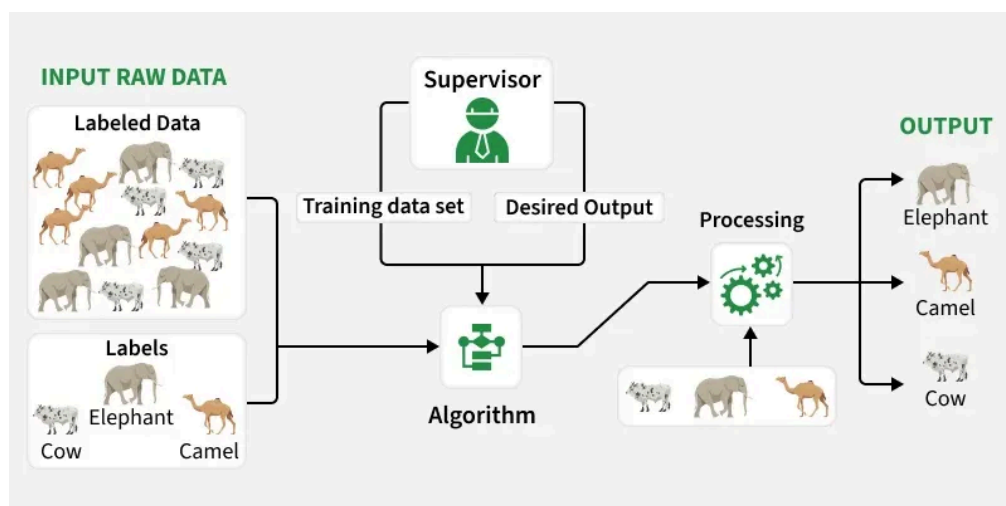
The model is trained using **labeled data** (input + correct output). The goal is to learn a mapping between input and output.

Types:

01. Classification (Yes/No, True/False)
02. Regression (continuous values)

Example:

01. Predicting house prices using past labeled data.
02. Email spam detection (spam / not spam).



Unsupervised Learning

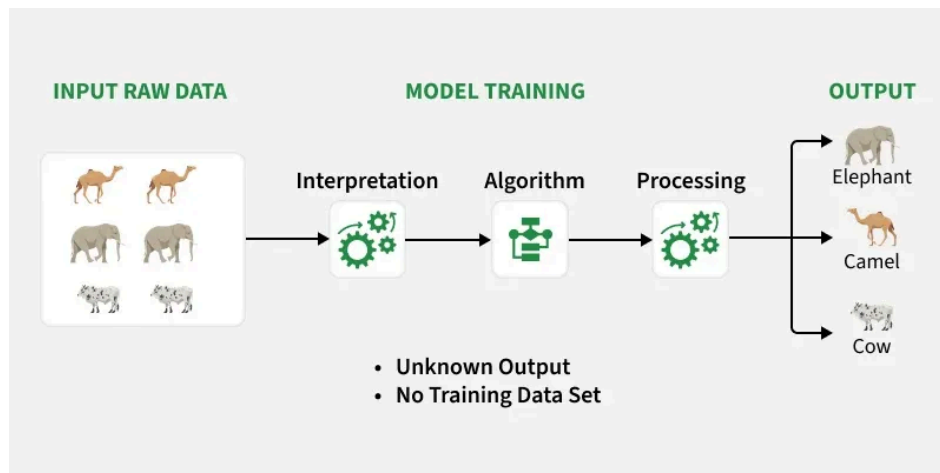
The model is trained using **unlabeled data** (only input, no output). The goal is to find hidden patterns or groups in data.

Types:

01. Clustering
02. Association

Example:

01. Grouping customers based on shopping behavior.
02. Finding similar movies for recommendation.



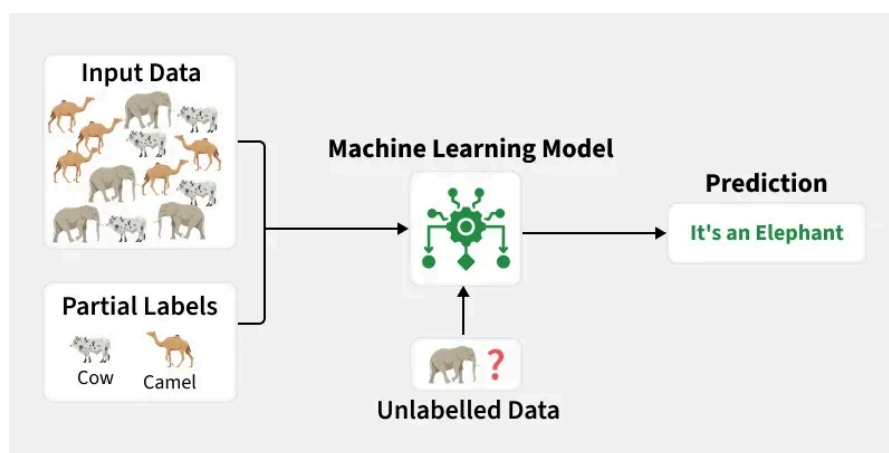
Semi-Supervised Learning

Uses a **small amount of labeled data** and a **large amount of unlabeled data**.

Combines supervised and unsupervised learning.

Example:

01. Image classification where only some images are labeled.
02. Speech recognition systems trained with limited labeled audio.



4.2 Supervised Learning

Learning a Class from Examples

The model learns by studying many examples with correct answers. It identifies patterns and applies them to new data.

Example:

Email dataset:

01. Spam $\rightarrow$ 1 02. Not Spam $\rightarrow$ 0

The model learns what makes an email spam or not spam.

Introduction of different types of Supervised Machine

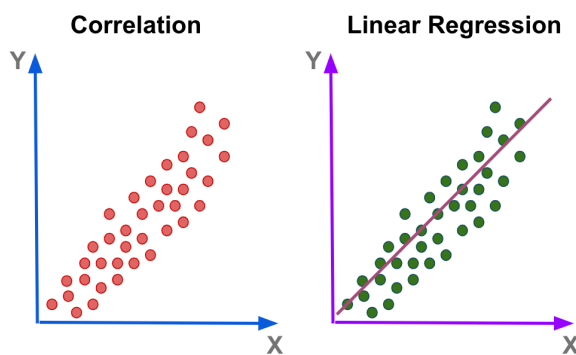
Learning Algorithms:

Linear Regression

Used for **predicting continuous values**. Finds a straight-line relationship between input and output.

Example:

- Predicting house prices based on size.

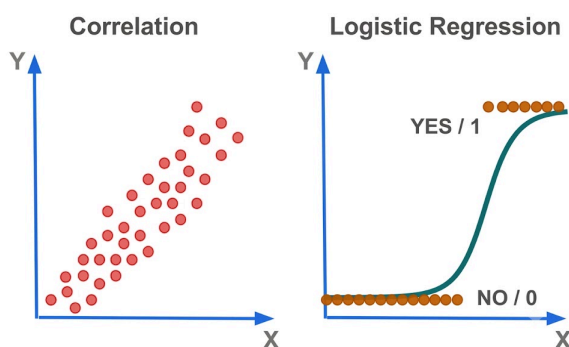


Logistic Regression

Used for **classification problems (yes/no, 0/1)**. Uses probability to decide class.

Example:

- Predicting whether a student will pass or fail.

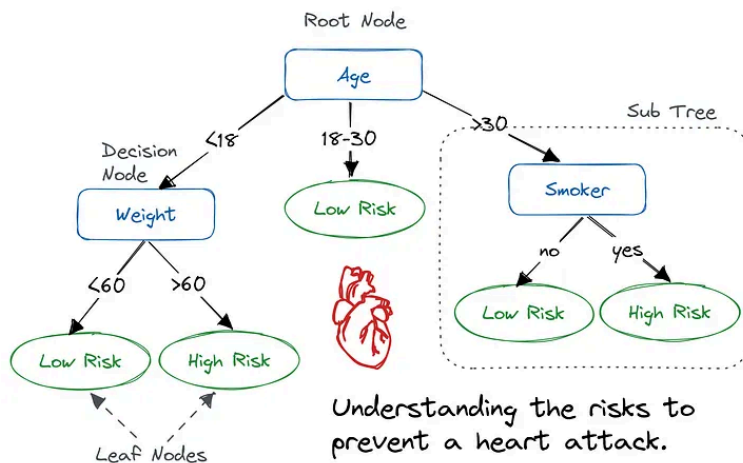


Decision Tree

A tree-like model that makes decisions based on conditions. Splits data step by step.

Example:

- Deciding whether to play cricket based on weather (sunny, rainy).

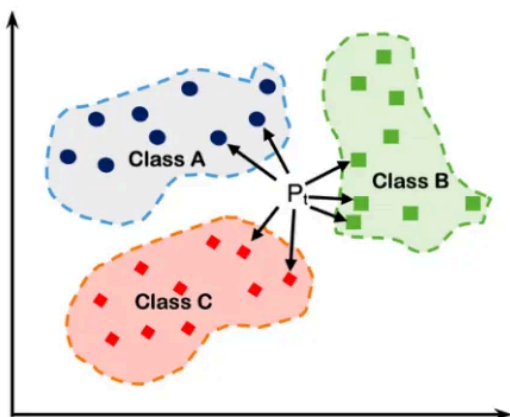


K-Nearest Neighbors (K-NN)

Classifies data based on **closest neighbors**. Chooses the majority class among nearest points.

Example:

- Classifying a new fruit based on similar nearby fruits.

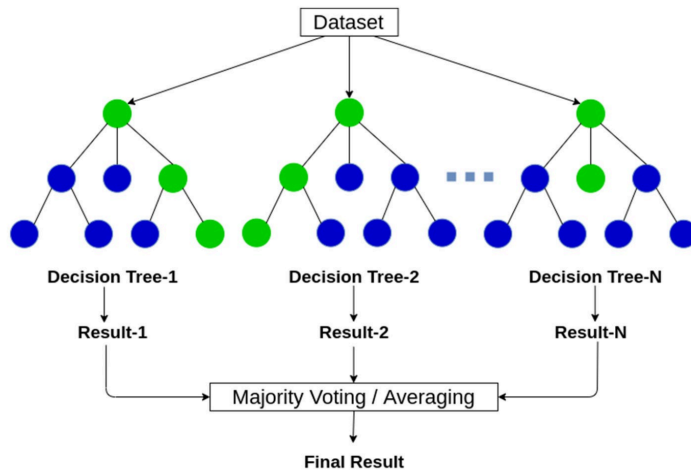


Random Forest

Collection (ensemble) of many decision trees. Final result is based on majority voting.

Example:

- Predicting loan approval using multiple decision trees.



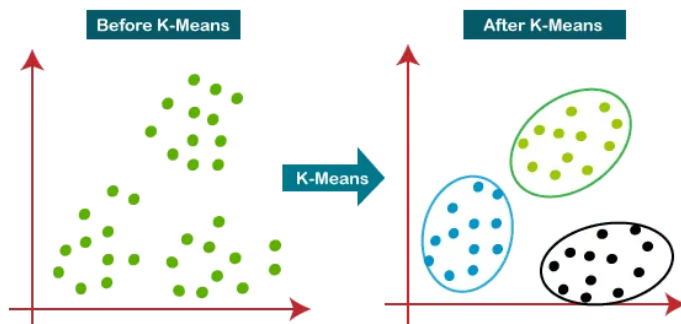
4.3 Unsupervised Learning: Introduction of different types of Unsupervised Learning Algorithm

K-Means Clustering

Groups data into **K clusters** based on similarity. Each data point belongs to the nearest cluster center (centroid).

Example:

- Grouping customers into 3 segments: low, medium, high spenders.

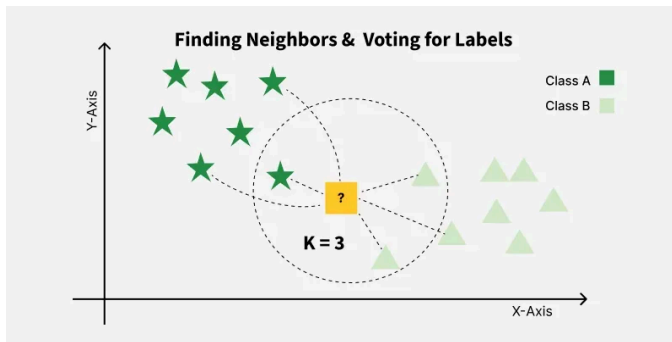


K-Nearest Neighbors (K-NN)

Finds the **closest data points (neighbors)** to classify or predict data. Uses distance (like Euclidean distance) to find similarity.

Example:

- Recommending products based on similar users' choices.

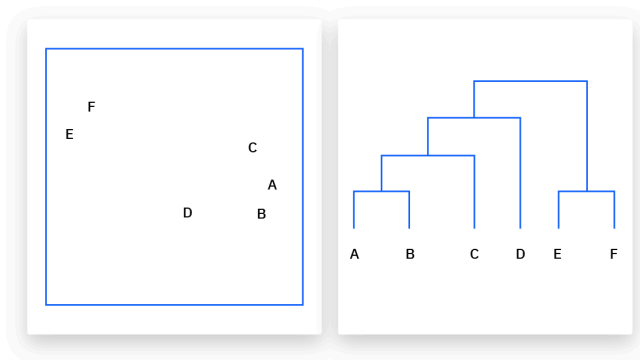


Hierarchical Clustering

Creates a **tree-like structure (dendrogram)** of clusters. Can be: Agglomerative (bottom-up) or Divisive (top-down)

Example:

- Organizing animals based on similarities (species grouping).

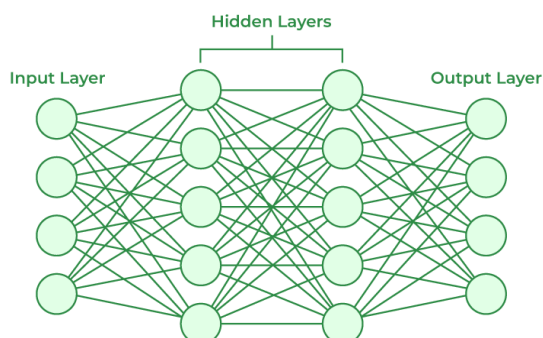


Neural Networks

Inspired by the human brain structure. Can learn complex patterns from data. Can be used in both supervised and unsupervised learning.

Example:

- Image recognition and pattern detection.



4.4 Model evaluation

Process of checking how well a machine learning model performs. Ensures the model works well on unseen data.

Cross-Validation

A technique used to test a model multiple times using different data splits. The dataset is divided into parts (folds). Model is trained on some folds and tested on the remaining fold. Process is repeated to get reliable results.

Example:

- 5-Fold Cross Validation:
 - Data is split into 5 parts
 - Model trains on 4 parts and tests on 1 part
 - Repeated 5 times

Benefits of Cross-Validation

Gives **more accurate performance results**. Reduces risk of overfitting. Uses data efficiently. Works well for small datasets.

Example:

- Instead of testing once, the model is tested multiple times for better accuracy.

Positive and Negative Class in Cross-Validation

Used in classification problems.

Positive Class

The class we are interested in detecting. Usually labeled as **1**.

Example: Spam email detection → Spam = Positive class

Negative Class

The opposite of positive class. Usually labeled as **0**.

Example: Spam email detection → Not Spam = Negative class